Qian Ouyang

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Education

Xi'an Jiaotong University 2023 - 2027 (expected)
B.Eng. in Artificial Intelligence
Qian Xuesen Honors College | Artificial Intelligence Experimental Class

University of Notre Dame August - December 2025
Undergraduate Exchange Student (non-degree) | Semester GPA: 3.917/4.0 (12 credits)
Coursework: Computer Vision (graduate, A); Natural Language Processing (A); Computer Graphics (A).

Research Experience

Undergraduate Research Assistant, University of Notre Dame October 2025 - Present
Advisor: Prof. Tengfei Luo (Aerospace and Mechanical Engineering) | iSURE Research Participant, Summer 2026

Machine Learning Potentials for Polymer Properties

- Developed workflows for finite-strain calculations with UMA (FAIR-Chem) and LAMMPS, using positive and negative strains in six modes and stress tensor postprocessing to estimate stiffness matrices and elastic moduli for individual and batched polymer structures.
- Combined classical structure preparation with MACE molecular dynamics to estimate polymer densities at 300 K, analyze trajectory fluctuations, and compare with reference values. Assessed simulation time requirements for thermal conductivity calculations.

Representation Learning for Molecular Dynamics

- Integrated pretrained MACE features, atomic velocities, and local geometry to predict displacements over short intervals in atomistic water. Compared temporal JEPA training with direct physical supervision, evaluating oxygen and hydrogen errors separately.
- Benchmarked JEPA predictors against persistence and linear autoregression in coarse-grained polystyrene, comparing representations of local packing and chain geometry for predicting chain size and local coordination.
- Tested the contribution of temporal information by comparing true and shuffled pairs of molecular configurations. Froze the encoder and adapter, then trained probes on the same downstream displacement prediction task.

Industry Experience

JIESHUN Technology January - March 2026
Deep Learning Algorithm Intern | Intelligent Algorithm Department | Shenzhen, China

- Reproduced and fine-tuned end-to-end OCR and visual sequence recognition models for license plates with varied layouts using Middle Eastern traffic data.
- Refined model architecture and training strategies, evaluated recognition across traffic scenes, and analyzed failure cases involving viewpoint changes, lighting variation, and cluttered backgrounds.

Technical Skills

Programming & Tools: Python, C++, PyTorch, Git, Linux Shell, Conda
Molecular Simulation: LAMMPS, MACE, FAIR-Chem (UMA)
HPC & Cloud: Notre Dame CRC (Grid Engine, HTCondor), AWS, CPU/GPU Batch Computing